# Calculus II

Solution to Assignment 6

## Part I. Regular Problems

1. Express each linear function as $f(\mathbf{X}) = \mathbf{C} \cdot \mathbf{X}$ for some vector $\mathbf{C}$ or $\mathbf{F}(\mathbf{X}) = \mathbf{C}\mathbf{X}$ for some matrix $\mathbf{C}$.

   (a) $f(x_1, x_2) = x_1 + 2x_2$

   $$f(\mathbf{X}) = \begin{bmatrix} 1 & 2 \end{bmatrix} \mathbf{X}$$

   (b) $f(x_1, x_2, x_3) = x_1 + 2x_2$

   $$f(\mathbf{X}) = \begin{bmatrix} 1 & 2 & 0 \end{bmatrix} \mathbf{X}$$

   (c) $\mathbf{F}(x_1, x_2) = (x_1, x_1 + 2x_2)$

   $$\mathbf{F}(\mathbf{X}) = \begin{bmatrix} 1 & 0 \\ 1 & 2 \end{bmatrix} \mathbf{X}$$

   (d) $\mathbf{F}(x_1, x_2, x_3) = (x_1, x_1 + 2x_2)$

   $$\mathbf{F}(\mathbf{X}) = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 0 \end{bmatrix} \mathbf{X}$$

   (e) $\mathbf{F}(x_1, x_2, x_3, x_4) = (x_4 - x_2, x_3 - x_1, x_2 + 5x_1, x_1 + x_2 + x_3 + x_4)$

   $$\mathbf{F}(\mathbf{X}) = \begin{bmatrix} 0 & -1 & 0 & 1 \\ -1 & 0 & 1 & 0 \\ 5 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix} \mathbf{X}$$

(f) $\mathbf{F}(x_1, x_2) = (x_1, 5x_1, -x_2, -2x_1, x_2)$

$$\mathbf{F}(\mathbf{X}) = \begin{bmatrix} 1 & 0 \\ 5 & 0 \\ 0 & -1 \\ -2 & 0 \\ 0 & 1 \end{bmatrix} \mathbf{X}$$

(g) $\mathbf{F}(x_1, x_2, x_3, x_4, x_5) = (x_1, 5x_1, -x_2, -2x_1, x_2)$

$$\mathbf{F}(\mathbf{X}) = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 5 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ -2 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \end{bmatrix} \mathbf{X}$$

2. The function

$$f(x, y) = 5 + x + x^2 + y^2$$

is the sum of the constant function 5, the linear function $x$, and the degree 2 polynomial $x^2 + y^2$.

(a) Give two examples of points $(x, y)$ near $(0, 0)$ for which the approximation by the first two parts,

$$f(x, y) \approx 5 + x$$

differs from $f(x, y)$ less than $\frac{1}{100}$.
Choose points near $(0, 0)$:

$$(x, y) = (0.05, 0.05), \quad (x, y) = (-0.05, 0.05)$$

For these points, the difference $f(x, y) - (5 + x)$ will be less than $\frac{1}{100}$:

$$|f(x, y) - (5 + x)| = x^2 + y^2 = 0.0025 + 0.0025 = 0.005 < \frac{1}{100}.$$

(b) Show that

$$g(u, v) = f(1 + u, 2 + v)$$

is the sum of a constant function, a linear function of $(u, v)$, and a degree 2 polynomial function of $u$ and $v$.

Expanding $g(u, v)$:

$$\begin{aligned} g(u,v) &= 5 + (1+u) + (1+u)^2 + (2+v)^2 \\ &= 11 + 3u + 4v + u^2 + v^2. \end{aligned}$$

This gives the sum of:
- Constant term: 5
- Linear term in $u$ and $v$: $3u + 4v$
- Degree 2 polynomial in $u$ and $v$: $u^2 + v^2$.

3. Let $f(\mathbf{X}) = \|\mathbf{X}\|^2$, $\mathbf{X}$ in $\mathbb{R}^4$. Let $\mathbf{A}$ be in $\mathbb{R}^4$ and define

$$g(\mathbf{X}) = \|\mathbf{A}\|^2 + 2\mathbf{A} \cdot (\mathbf{X} - \mathbf{A})$$

(a) Let $\mathbf{U} = \mathbf{X} - \mathbf{A}$. Use the formula

$$\|\mathbf{A} + \mathbf{U}\|^2 = \|\mathbf{A}\|^2 + 2\mathbf{A} \cdot \mathbf{U} + \|\mathbf{U}\|^2$$

to show that the difference between $f(\mathbf{X})$ and $g(\mathbf{X})$ is $\|\mathbf{U}\|^2$.
Using the formula, we get that:

$$\begin{aligned} f(\mathbf{X}) - g(\mathbf{X}) &= \|\mathbf{X}\|^2 - \|\mathbf{A}\|^2 - 2\mathbf{A} \cdot \mathbf{U} \\ &= \|\mathbf{A} + \mathbf{U}\|^2 - \|\mathbf{A}\|^2 - 2\mathbf{A} \cdot \mathbf{U} \\ &= \|\mathbf{U}\|^2 \end{aligned}$$

(b) Show that the difference between $f(\mathbf{X})$ and $g(\mathbf{X})$ does not exceed $10^{-4}$ when $\|\mathbf{X} - \mathbf{A}\| < 10^{-2}$.
Since $\|\mathbf{U}\|^2 < (10^{-2})^2 = 10^{-4}$, the difference is less than $10^{-4}$.

4. Sketch or describe the $c$ level sets, $f(\mathbf{X}) = c$ in $\mathbb{R}^3$, for the following functions $f$ from $\mathbb{R}^3$ to $\mathbb{R}$ and values $c$.

(a) $f(\mathbf{X}) = \|\mathbf{X}\|^2$, $c = \frac{1}{2}, 1, 2$
The level sets are spheres centered at the origin with radii:

$$c = \frac{1}{2}: \quad \text{radius} = \sqrt{\frac{1}{2}}; \quad c = 1: \quad \text{radius} = 1; \quad c = 2: \quad \text{radius} = \sqrt{2}$$

(b) $f(\mathbf{X}) = \|\mathbf{X}\|$, $c = \frac{1}{2}, 1, 2$
The level sets are spheres centered at the origin with radii:

$$c = \frac{1}{2}: \quad \text{radius} = \frac{1}{2}; \quad c = 1: \quad \text{radius} = 1; \quad c = 2: \quad \text{radius} = 2$$

(c) $f(\mathbf{X}) = \frac{1}{\|\mathbf{X}\|^2}$, $c = \frac{1}{2}, 1, 2$

The level sets are spheres centered at the origin with radii:

$$c = \frac{1}{2}: \quad \text{radius} = \sqrt{2}; \quad c = 1: \quad \text{radius} = 1; \quad c = 2: \quad \text{radius} = \frac{1}{\sqrt{2}}$$

5. The slope of the graph of $\cos(cx)$ lies in the interval $[-c, c]$. Fill in the missing numbers.

**Solution.** Set $f(x) = \cos(cx)$. By the mean value theorem:

$$|\cos(cx) - \cos(ca)| = |f'(\tilde{x})(x - a)| \leq |c| \cdot |x - a|,$$

where $\tilde{x}$ is between $x$ and $a$. Therefore:

(a) If $|x - a| < \frac{\epsilon}{2}$, then $|\cos(2x) - \cos(2a)| < \epsilon$.

(b) If $|y - b| < \delta$, then $|\cos(3y) - \cos(3b)| < 3\delta$.

(c) Since

$$\begin{aligned}
&|\cos(2x)\cos(3y) - \cos(2a)\cos(3b)| \\
&\quad = |\cos(2x)\cos(3y) - \cos(2x)\cos(3b) \\
&\qquad + \cos(2x)\cos(3b) - \cos(2a)\cos(3b)| \\
&\quad \leq |\cos(2x)| \cdot |\cos(3y) - \cos(3b)| \\
&\qquad + |\cos(3b)| \cdot |\cos(2x) - \cos(2a)| \\
&\quad \leq |\cos(3y) - \cos(3b)| + |\cos(2x) - \cos(2a)| \\
&\quad \leq 3|y - b| + 2|x - a| \\
&\quad \leq 3|y - b| + 3|x - a| \\
&\quad \leq \sqrt{3^2(x - a)^2 + 3^2(y - b)^2} + \sqrt{3^2(x - a)^2 + 3^2(y - b)^2} \\
&\quad = 6\|(x, y) - (a, b)\|,
\end{aligned}$$

if $\|(x, y) - (a, b)\| < \frac{\epsilon}{6}$, then

$$|\cos(2x)\cos(3y) - \cos(2a)\cos(3b)| < \epsilon$$

6. Show that the function

$$f(x, y, z) = \frac{\sin(x^2 + y^2)}{e^{z+y}}$$

is continuous at all points $(x, y, z)$.

**Proof.** We first show the following claim: If $f : D \subset \mathbb{R}^m \to \mathbb{R}$ is continuous on $D$, $g : A \subset \mathbb{R} \to \mathbb{R}$ is continuous on $A$, and the range of $f$ is contained in $A$, then the composition $g \circ f : D \to \mathbb{R}$ is continuous on $D$.

Let $\mathbf{X}$ be in $D$.

- Since $g$ is continuous at $f(\mathbf{X})$, for every $\epsilon > 0$, there exists $\delta > 0$ such that:

$$\|f(\mathbf{X}) - y\| < \delta \Rightarrow |g(f(\mathbf{X})) - g(y)| < \epsilon.$$

- Since $f$ is continuous at $\mathbf{X}$, there exists $\gamma > 0$ such that:

$$\|\mathbf{X} - \mathbf{Z}\| < \gamma \Rightarrow \|f(\mathbf{X}) - f(\mathbf{Z})\| < \delta.$$

- Combining these, for $\|\mathbf{X} - \mathbf{Z}\| < \gamma$, we have:

$$|g(f(\mathbf{X})) - g(f(\mathbf{Z}))| < \epsilon.$$

- Therefore, $g \circ f$ is continuous at $\mathbf{X}$.

According to the proposition that we have proved, both $\sin(x^2 + y^2)$ and $e^{z+y}$ are continuous everywhere, since they are compositions of continuous functions. Moreover,

$$e^{z+y} \neq 0, \forall x, y, z \in \mathbb{R},$$

which implies that

$$\frac{\sin(x^2 + y^2)}{e^{z+y}}$$

are continuous everywhere. Here we applied the theorem that quotients of continuous functions are again continuous where the denominator is not equal to 0.

## Part II. Group Problems

1. Sketch or describe the $c$ level sets, $f(x, y) = c$ in $\mathbb{R}^2$, for the following functions $f$ and values $c$.

   (a) $f(x, y) = x + 2y$, $c = -1, 0, 1, 2$
   The level sets are straight lines given by:

$$y = -\frac{1}{2}x + \frac{1}{2}c$$

   (b) $f(x, y) = xy$, $c = -1, 0, 1, 2$
   For $c = -1, 1, 2$, the level sets are hyperbolas, and the equations are:

$$y = c/x, \quad x \neq 0;$$

   for $c = 0$, the level sets are two straight lines:

$$x = 0 \quad \text{and} \quad y = 0.$$

(c) $f(x, y) = x^2 - y$, $c = -1, 0, 1, 2$
The level sets are parabolas, given by:

$$y = x^2 - c$$

2. Determine the set of points at which the function is continuous.

**Solution.** Note that in Problem 6 we have shown that compositions of continuous functions are again continuous.

(a) $f(x, y) = \frac{xy}{1+e^{x-y}}$
The functions $xy$ and $1 + e^{x-y}$ are continuous everywhere, and $1 + e^{x-y}$ is never zero, so $f(x, y) = \frac{xy}{1+e^{x-y}}$ is continuous on its domain $\mathbb{R}^2$.

(b) $f(x, y) = \frac{1+x^2+y^2}{1-x^2-y^2}$
Since the numerator and denominator (as products of linear functions) are both continuous functions on $\mathbb{R}^2$, $f$ is continuous on its domain

$$\left\{(x, y) \mid 1 - x^2 - y^2 \neq 0\right\} = \left\{(x, y) \mid x^2 + y^2 \neq 1\right\}.$$

(c) $f(x, y, z) = \arcsin(x^2 + y^2 + z^2)$
$f(x, y, z) = h(g(x, y, z))$ where $g(x, y, z) = x^2+y^2+z^2$, a polynomial that is continuous everywhere, and $h(t) = \arcsin t$, continuous on $[-1, 1]$. Thus, $f$ is continuous on its domain $\left\{(x, y, z) \mid -1 \leq x^2 + y^2 + z^2 \leq 1\right\} = \left\{(x, y, z) \mid x^2 + y^2 + z^2 \leq 1\right\}$, so $f$ is continuous on the unit ball.